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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

Assuming $3x^3 - 4y^2 + 2xy = 0$ defines y as a differentiable function of x , use the theorem $\frac{dy}{dx} = -\frac{F_x}{F_y}$ to find $\frac{dy}{dx}$ at the point $(2, -2)$.

According to the theorem given above, $\frac{dy}{dx} = -\frac{F_x}{F_y}$ where $F_x = \frac{\partial F}{\partial x}$ and $F_y = \frac{\partial F}{\partial y}$. Therefore, to find $\frac{dy}{dx}$, find each partial derivative separately and substitute them into the relation given in the theorem.

Differentiate $F(x,y) = 3x^3 - 4y^2 + 2xy$ with respect to x . Treat y as a constant.

$$\begin{aligned} F_x &= \frac{\partial}{\partial x} (3x^3 - 4y^2 + 2xy) \\ &= 9x^2 + 2y \end{aligned}$$

Differentiate $F(x,y) = 3x^3 - 4y^2 + 2xy$ with respect to y . Treat x as a constant.

$$\begin{aligned} F_y &= \frac{\partial}{\partial y} (3x^3 - 4y^2 + 2xy) \\ &= -8y + 2x \end{aligned}$$

Write $\frac{dy}{dx}$ using the theorem $\frac{dy}{dx} = -\frac{F_x}{F_y}$.

$$\frac{dy}{dx} = -\frac{9x^2 + 2y}{-8y + 2x}$$

Substitute $x = 2$ and $y = -2$ and evaluate to find $\frac{dy}{dx}$ at the given point.

$$\begin{aligned} \left. \frac{dy}{dx} \right|_{(2, -2)} &= -\frac{9(2)^2 + 2(-2)}{-8(-2) + 2(2)} \\ &= -\frac{32}{20} \\ &= -\frac{8}{5} \end{aligned}$$